MAT110 - Differential Calculus and Co-ordinate Geometry, Spring 2026
Assignment - 01

1. Explain in your own words what is meant by the equation

$$\lim_{x \to 2} f(x) = 5$$

Is it possible for this statement to be true and yet $f(2) = 3$? Explain. [1]

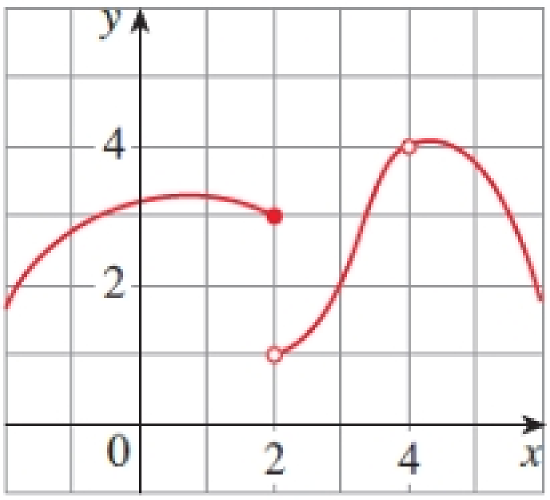
2. Explain what it means to say that

$$\lim_{x \to 1^-} f(x) = 3 \quad \text{and} \quad \lim_{x \to 1^+} f(x) = 7$$

Here, is it possible for $\lim_{x \to 1} f(x)$ to exist? Explain. [1]

3. Use the graph of f to state the value of each quantity, if it exists. If it does not exist, explain why. [3]

(a) $\lim_{x \to 2^-} f(x)$
(b) $\lim_{x \to 2^+} f(x)$
(c) $\lim_{x \to 2} f(x)$
(d) $f(2)$
(e) $\lim_{x \to 4} f(x)$
(f) $f(4)$



4. Evaluate the limit, if it exists. [3]

(a) $\lim_{x \to -1} \dfrac{x+1}{x^3+1}$

(b) $\lim_{x \to 0} \dfrac{\sqrt{9+x}-3}{x}$

(c) $\lim_{h \to 0} \frac{(x+h)^3 - x^3}{h}$

(d) $\lim_{x \to 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$

(e) $\lim_{x \to 0} \left(\frac{1}{x} - \frac{1}{x^2+x} \right)$

(f) $\lim_{x \to -4} \frac{\sqrt{x^2+9} - 5}{x+4}$

5. Evaluate the limit, if it exists. [3]

(a) $\lim_{x \to -\infty} \frac{(x^2+1)(2x^2-1)}{(x^2+2)^2}$

(b) $\lim_{x \to -\infty} \frac{3x^2+x}{x^3-4x+1}$

(c) $\lim_{x \to \infty} \frac{\sqrt{x+3x^2}}{4x-1}$

(d) $\lim_{x \to -\infty} \left(\sqrt{4x^2+3x} + 2x \right)$

(e) $\lim_{x \to \infty} \frac{\sin^2 x}{1+x^2}$

(f) $\lim_{x \to \infty} (e^{-x} + 2\cos 3x)$

6. Are the following functions continuous at the given point? If not, explain why. [2]

(a)

$$f(x) = \begin{cases} \cos x, & x < 0 \\ 0, & x = 0 \\ 1 - x^2, & x > 0 \end{cases}$$

at $x = 0$

(b)

$$f(x) = \begin{cases} 1 - x^2, & x \leq 1 \\ \ln x, & x > 1 \end{cases}$$

at $x = 1$

7. (a) Find the values of a and b that make $f(x)$ continuous everywhere. [1]

$$f(x) = \begin{cases} \dfrac{x^2-4}{x-2}, & x < 2 \\ ax^2 - bx + 3, & 2 \leq x < 3 \\ 2x - a + b, & x \geq 3 \end{cases}$$

(b) Find the value of x for which the function

$$f(x) = \frac{x^2 - 9}{x - 3}$$

is not continuous. Is the discontinuity removable or not? Explain. [1]

8. Find $\dfrac{dy}{dx}$ for: [6]

(a)

$$y = \sin\left(2\tan^{-1}\sqrt{\frac{1-x}{1+x}}\right)$$

(b)

$$y = \ln\sqrt{\frac{1+\sin x}{1-\sin x}}$$

(c)

$$y = \ln\left(\frac{1}{\sqrt{\cos x}}\right)$$

(d)

$$x = a\sec^2\theta, \qquad y = a\tan^2\theta$$

(e)

$$x^3 + y^3 + 4x^2y - 25 = 0$$

(f)

$$(\sin x)^{\cos x} + (\cos x)^{\sin x}$$

9. Find (a) the intervals on which f is increasing, (b) the intervals on which f is decreasing, (c) the intervals on which f is concave up, (d) the intervals on which f is concave down, (e) the x-coordinate of all inflection points and critical points and (f) relative maxima and minima. [6]

(i) $f(x) = x^4 - 8x^2 + 16$

(ii) $f(x) = \dfrac{x^2}{x^2 + 2}$

(iii) $f(x) = \sqrt[3]{x + 2}$

10. Answer the followings: [3]

(a) Verify the hypothesis of Rolle's Theorem for the following functions:
$f(x) = x^2 - 6x + 8; \quad [2, 4]$

(b) Verify the hypothesis of Mean Value Theorem for the following functions:
$f(x) = x^3 + x - 4; \quad [-1, 2]$

[Total Marks = 30]

–Best of luck–